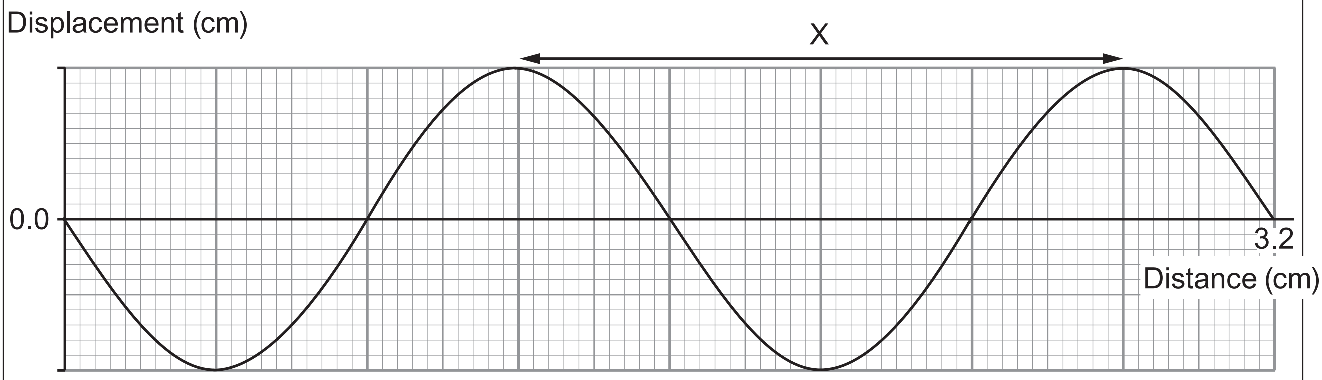
Examiner only

2. A teacher demonstrates water waves in a ripple tank.

(a) The diagram represents the waves produced at a particular frequency.



(i) Use the information in the diagram to calculate the distance X. [2]

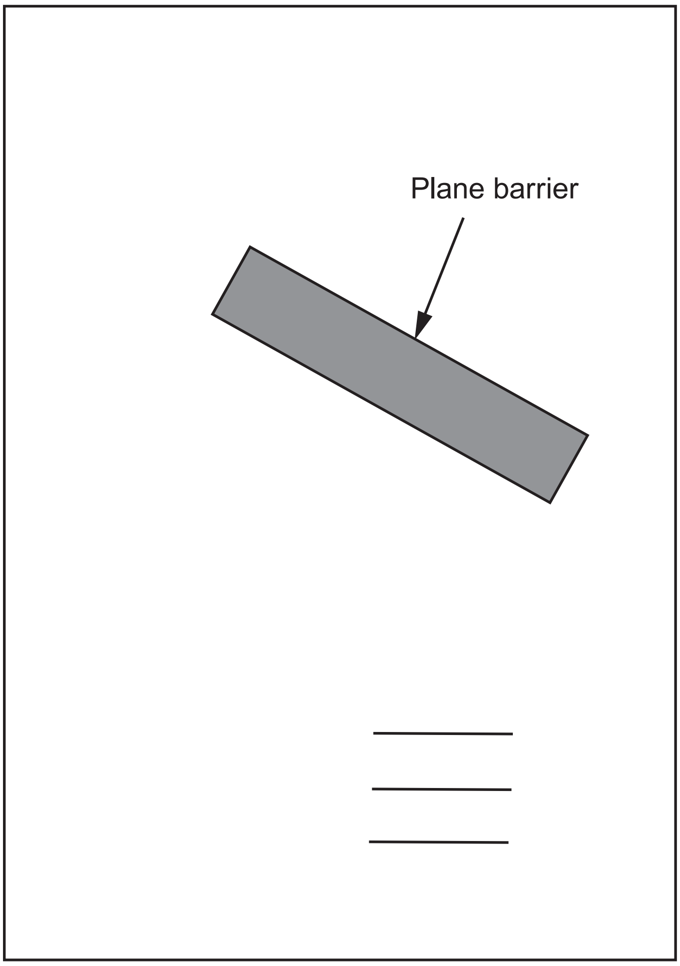
X = ... cm

(ii) A wave takes 5 s to travel 40 cm along the surface of the water in the ripple tank. Use equations from page 2 and the information above to calculate the frequency of the wave. [4]

Frequency = ... Hz

Examiner only

(b) Wavefronts travel towards a plane barrier. **Complete the diagram** to show the path of the wavefronts as they are incident on and reflect off the barrier. [2]



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